

One fitted parameter restores the tested voltage margin



Fresh plant,
nominal model
within bounds

--- 18.5 min



High resistance,
stale model
voltage bound crossed

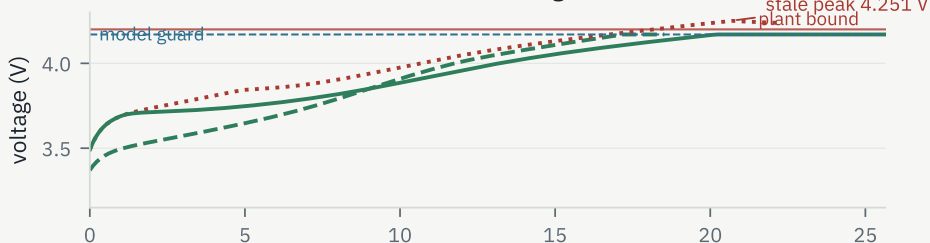
... 22.3 min



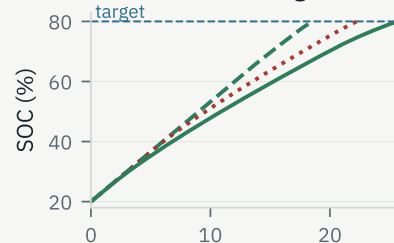
High resistance,
fitted model
within bounds

— 25.6 min

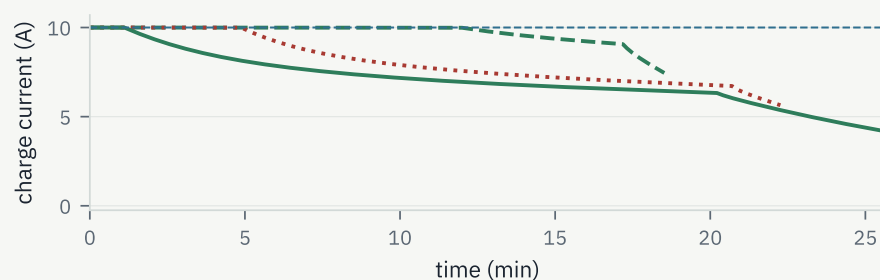
Terminal voltage



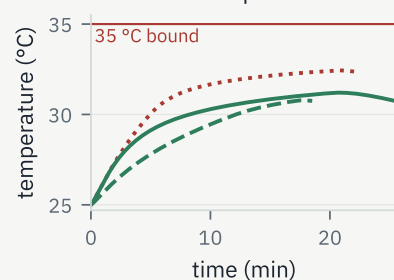
Stored charge



Governor action



Cell temperature



plant and controller model	target (min)	peak V	time >4.20 V (s)	peak cell °C
Fresh plant, nominal model	18.5	4.170	0	30.8
High resistance, stale model	22.3	4.251	246	32.4
High resistance, fitted model	25.6	4.170	0	31.2